# The Artificial Neuron

Name _____

Work with a partner. Compare answers when you finish each problem.

**Word bank:** adds compares multiplies

**1. Should I go?** Answer 1 (yes) or 0 (no). Then choose how much each one matters to you (0 to 3).

| | **Answer (1 or 0)** | **How much it matters** | **Answer $\times$ how much** |
|---|---|---|---|
| Is it sunny? | | | |
| Are my friends going? | | | |
| Is my homework done? | | | |
| | | **Total** | |

Is the total 4 or more? _____ Do I go? _____ Did my partner decide the same? _____

**2. Multiply, add, compare.**

An artificial neuron _____ each input by its weight, _____ them up, and _____ the total with the threshold.

Inputs: 1, 0, 1. Weights: 3, 2, $-1$. Threshold: 2.

Total = $(1 \times 3) + (0 \times 2) + (1 \times -1)$ = _____ Does it fire (2 or more)? _____

Inputs: 1, 1, 1. Weights: 1, $-2$, $-1$. Threshold: 2.

Total = _____ Does it fire? _____

**3. Be a neuron.** Weights: **A:** +2, **B:** +2, **C:** $-1$. Threshold: **3 or more**.

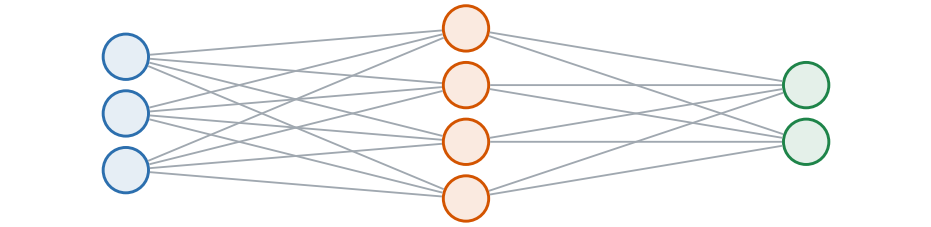
| **Round** | **Inputs that are on** | **Total** | **Fire?** |
|---|---|---|---|
| 1 | A and B | | |
| 2 | A and C | | |
| 3 | A, B, and C | | |
| 4 | B and C | | |
| 5 | A only | | |

**Word bank:** hidden input output threshold weight
*You can use a word more than once.*

**4. Real neuron, artificial neuron.** Match each part.

| Real neuron | Artificial neuron |
|---|---|
| Dendrites take signals in | |
| Strong or weak synapses | |
| Fires only if the total is big enough | |

**5. From one neuron to a network.** Name each layer.



_____ layer _____ layer _____ layer

Each line is a _____. How many lines? $3 \times 4 + 4 \times 2 =$ _____

Why is the middle layer called hidden? _____

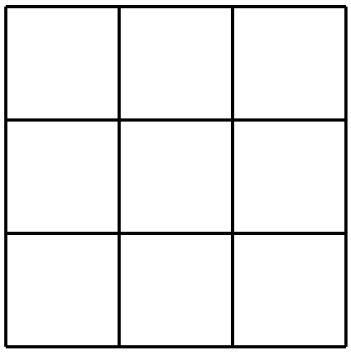
**6. Talk about it.** ChatGPT has billions of weights. Nobody can choose them by hand.

I think an AI with billions of weights gets them by _____

_____

_____

**Go Further (Day 2).** A neuron that sees. Write the weights that got the most test pictures right.



Threshold: _____

Test pictures right: _____ out of 6

Which squares got **positive** weights? Why? _____

Which squares got **negative** weights? Why? _____